

Education

The University of Chicago

2030

Bachelor of Science, Double Major in Computer Science & Computational and Applied Mathematics (CAM)

Relevant course work (2026-27): CMSC 14100 (placed out), CMSC 14200, CMSC 14300, MATH 15300 (Calculus III), MATH 15910 (Intro. to Proofs in Analysis), MATH 20300 (Real Analysis I)

St. Margaret’s Episcopal School

2022 – 2026

Diploma

GPA: 3.99 UW / 4.65 W | Salutatorian

Honors: National Latin Exam Gold (1x) & Silver (2x)

Projects

Pieta AI

August 2026

- Built and deployed a full-stack web app that analyzes and classifies a poem's sentiment with Next.js, React, Vercel, and Supabase.
- Fine-tuned and pre-trained a BERT encoder on 6k+ texts for a 4-class sentiment task (no impact/positive/negative/mixed) using Hugging Face's Trainer API.
- Built the evaluation pipeline, wrote custom metrics function, reaching ~86% accuracy and a 0.71 F1-score.
- Fine-tuned a second model via LoRA for poem sentiment classification.

Fraud Detector Model with Imbalanced Dataset

July 2026

- Built a 3-layer neural network with PyTorch to detect fraudulent transactions in a severely imbalanced dataset (~284,800 transactions, only 0.17% fraudulent).
- Manipulated data with Pandas and NumPy to create custom features for the model.
- Trained with BCEWithLogitsLoss and Adam, evaluating via precision-recall curves rather than accuracy given the class imbalance.
- Achieved 80% precision and 75% recall on the fraud class.

Data Analysis on Netflix User Activity Dataset

July 2026

- Cleaned a dataset using Pandas containing several csv files with missing values, un-aligned indices, and duplicate entries.
- Engineered features for a potential binary classification model predicting churn status of Netflix users.

Professional Experience

Intern | NASA, Goddard Space Flight Center (GSFC)

Summer 2024

- Tasked with analyzing carbon datasets for NASA scientists and grad students modeling climate change's long-term economic impacts.
- Assigned to clean and structure raw carbon data in Excel into model-ready inputs for the team's research.

NESST Intern | Smithsonian Institution

Summer 2023

- Integrated the UN's 17 Sustainable Development Goals into Smithsonian K-12 STEM curricula with Smithsonian researchers and global experts.
- Coordinated a socio-scientific framework for the Global Goals project, aligning regional education with international standards.
- Independently produced an animated climate-economics film as a final project.

Awards

- At 15, won the United States Academic Decathlon Individual Overall National Championship, Honors Division I (GPA 3.8+), then defended the title in 2024 for back-to-back national championships.
 - 2024: 1st nationally in 7 of 8 subjects (Economics, Literature, Mathematics, Science, Social Science, Music, Essay), 2nd in Art.
 - 2023: 1st nationally in 5 of 8 subjects; 2nd in Science, 3rd in Mathematics.
- John Locke Global Essay Competition Finalist (2024): finalist in the Economics category out of 36,000+ entries from 100+ countries. Essay topic: challenging the societal and economic impact of technology titans vs. drug cartels.

Skills

Programming languages: PyTorch, TensorFlow, Python, Java, Pandas, NumPy, Matplotlib, Seaborn, SQL, HTML, CSS
Computer software/ frameworks: Git, GitHub

Languages

English, Latin